

Experiment 10

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Title: Transaction Control in SQL.

1. Aim:

- a. To implement transaction control mechanisms in PostgreSQL to ensure data integrity and consistency by enforcing ACID properties.

2. Software Requirements:

- a. PostgreSQL (Database Server)
- b. pgAdmin
- c. OS: Windows

3. Objective:

- a. Data integrity: ensure that a system failure occurs during a transaction, the entire operation is rolled back, preventing partial corrupted data.
- b. Consistency: Maintains data validity by preventing scenarios like lost updates or inconsistent analysis.
- c. Isolation: Allows multiple transactions to occur simultaneously without interfering with each other.
- d. Recovery: Enables recovery of changed resources after a system failure.
- e. Error Management (Savepoints): Allows partial rollbacks using savepoints without reversing the entire transaction.

4. Procedure of the Practical:

- a. Start the system.
- b. Open pgAdmin.
- c. Create and select the database in which you want to perform the experiment.
- d. Establish connection to the database using Alt+Shift+Q.
- e. Run the following queries given in Experiment Steps (5).

Base Table

Payroll:

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	1	Amit	30000.00
2	2	Neha	40000.00
3	3	Ravi	50000.00

5. Experiment Steps:

```
BEGIN;
-- Update 1
UPDATE Payroll
SET salary = salary + 5000
WHERE emp_id = 1;
```

```
-- Savepoint
SAVEPOINT sp1;
```

```
-- Update 2
UPDATE Payroll
SET salary = salary + 7000
WHERE emp_id = 2;
```

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	3	Ravi	50000.00
2	1	Amit	35000.00
3	2	Neha	47000.00

```
-- Error simulation
UPDATE Payroll
SET salary = -1000
WHERE emp_id = 3;
```

```
ERROR: new row for relation "payroll" violates check constraint "payroll_salary_check"
Failing row contains (3, Ravi, -1000.00).
```

```
SQL state: 23514
Detail: Failing row contains (3, Ravi, -1000.00).
```

```
-- Rollback to savepoint
ROLLBACK TO sp1;
```

```
-- Commit valid changes
COMMIT;
```

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	2	Neha	40000.00
2	3	Ravi	50000.00
3	1	Amit	35000.00

6. Input/Output Details:

- The input for the experiment is the SQL queries mentioned in Experiment Steps (5).
- The output for each query is attached after the query itself.

7. Learning Outcome

- Understanding of transactions in SQL.
- Transaction Control Language commands.
- Ensuring ACID properties using transactions.